



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 29.07.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : MPBA G521/BITS F455
Course Title : Analytics for Supply Chain
Instructor-in-Charge : Dr Satyendra Kumar Sharma
Instructor(s) :
Tutorial/Practical Instructors:

1. Course Description:

Supply chain performance, Descriptive analytics, Supply chain forecasting, studying holt, winter and ARIMA models, Supply chain segmentation. Single period and multi-period inventory modeling, Multi-echelon inventory models, Network flow models, Distribution planning, Sales, & Operations planning, Vehicle routing problems, Simulation in supply chain risk assessment

2. Scope and Objective of the Course:

Supply chain analytics is amalgamation of supply chain management and data analytics. In this course you would be given basics and tool/techniques of data analytics with the domain of SCM. Students will be exposed to fundamental theories in business practice, and limitations and difficulties of supply chain management theories in solving real problems. The basic objective of this course is to help students develop the skill found in competent supply chain manager as well as analyst.

3. Text Books:

- T1 Supply chain Analytics by T.A.S. Vijayraghvan by Wiley
- T2 Supply chain management by Sunil Chopra Pearson Education

4. Reference Books:

- R1. Supply Chain Analytics: Beginner's Guide Gerard Blokdyk CreateSpace IndependentPublishing Platform, 07-Oct-201
- R2. Supply Chain Network Design: Applying Optimization and Analytics to the ... By Michael Watson, Sara Lewis, Peter Cacioppi, Jay Jayaraman
- R3. Supply Chain Optimization Through Segmentation and Analytics Book by Gerhard Plenert
- R4. Business forecasting



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5. Course Plan:

Learning Outcomes:

No	Learning Outcomes
LO1	Able to understand how various types of analytics can be used to build supply chain models for better decision making.
LO2	Compare and distinguish various supply chain metrics used for measuring supply chain performance. How the SCOR model can be used for SCP
LO3	Apply and creating dashboards for real time performance of the supply chain.
LO4	Using the forecasting tools/techniques to make good use of forecasts for demand planning
LO5	Understand the supply chain network design concepts and making supply chain network design models.
LO6	Be able to evaluate different demand scenarios to netter plan sales and operations planning.
LO7	Develop and demonstrate solutions to supply chain segmentation.
LO8	Analyze/ Evaluate real life supply chain situations and identify/ apply the appropriate analytics, tools/techniques to solve it in a optimal/ effective manner.



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Lecture Plan

Lecture No.	Learning Objectives	Topics to be covered	TB Chapter No.
1-2	Introduction to supply chain analytics	Importance of supply chain analytics and its types and various examples	T 1 chapter no.1 R1 chapter no 1
3-4	Data bases, Supply chain data bases	Relational databases, Creating supply chain data bases	T1 Chapter 6
5-6	Supply chain performance Descriptive analytics	Understanding SC key performance indicators, finding top KPI	T1 Chapter no.2
7-8	Introduction of Optimization modelling	Linear programming in Excel, Sensitivity analysis, Hub location models	T2 Chapter 12, 13
9-11	Facility location and Discrete location models	Center of Gravity models, Set covering models, maximal covering problem, Capacitated fixed Charge models	T2 Chapter 3
12-13	Network flow models	Shortest path problem, maximum flow and minimums cost model	T1 Chapter no. 4 and 5
14-16	Network decision models under uncertainty	SC Risks, Decision tree models	
17-19	Tactical and Operational decision making in warehousing	Space determination, layout decision, allocating products to storage locations	T2 Chapter 3
20	Supply chain forecasting, studying holt, winter models	Judgmental methods, Time series forecasting, Causal methods	R4 Chapter no. 9 and 10
20-21	Supply chain forecasting ARIMA models	Box Jenkin Methodology, Different ARIMA Models	R4 Chapter no. 9 and 10
21-23	Basic inventory models	P Model, Q Model, Quality discount models, Order aggregation models	T2 Chapter 6
24-25	Multi echelon network optimization	Extending inventory theories to multi Stage	T1 chapter no. 9, 10 and class notes
26-27	Goal Programming	Deviation variables, Setting up GP Model in Excel	
28-30	Supply chain sales and operations planning	Integrating marketing and supply chain models, Resilient S & OP Models	T1 chapter no 8
31-33	Supply chain segmentation	How the supply chain be designed for maximizing revenues	R3
34-37	Vehicle routing problems	Travelling salesman problem and vehicle routing problems: exact solutions and herustucs	T1 chapter no 14
38-40	Simulation and SC models	Basics of simulation and its types, Discrete event simulation, Monte Carlo simulation, Warranty cost model, What if analysis	T1 chapter 14



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Experiential Learning Components:

1. This course will feature experiential learning components in the form of projects and assignments (such as forecasting modeling, simulation, etc.) that are designed to enable the participants to learn by doing. This will also form part of the Evaluation Components for the course.
2. Please note that experiential learning components are integrated into the following lectures:

Lecture No.	Topic	Experiential Learning Component
5	Forecasting	MS Excel
6	ARIMA Modelling	MINITAB
7	Network flow models	Solver Excel
12	Principal Component analysis	SPSS
13	Cluster Analysis	SPSS
15	Simulation Model	What if Analysis

3. We will be building supply chain models using different software packages, hence students need a laptop/ PC preloaded with Microsoft Excel/SPSS/MINITAB for facilitating experiential learning.

Evaluation Scheme:

Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

No	Name	Type	Duration	Weight	Day, Date, Session, Time
1	Case study participation	Offline	-	15%	
	Quiz	Offline		5%	
	Course Project			15%	
2	Class participation			10%	
3	Mid-Semester Test	Closed Book	1.5 hours	25%	
	Comprehensive Exam	Open Book	3 hours	35%	

Note - Evaluation components can be tailored depending on the proposed model.

Important Information:

1. For Closed Book tests: No books or reference material of any kind will be permitted. Laptops/Mobiles of any kind are not allowed. Exchange of any material is not allowed.
2. For Open Book exams: Use of prescribed and reference text books, in original (not photocopies) is permitted. Class notes/slides as reference material in filed or bound form is permitted. However, loose sheets of paper will not be allowed. Use of calculators is permitted in all exams. Laptops/Mobiles of any kind are not allowed. Exchange of any material is not allowed.
3. If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student



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should follow the procedure to apply for the Make-Up Test/Exam. The genuineness of the reason for absence in the Regular Exam shall be assessed prior to giving permission to appear for the Make-up Exam. Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend the lectures, and take all the prescribed evaluation components such as Assignment/Quiz, Mid-Semester Test and Comprehensive Exam according to the evaluation scheme provided in the handout.